OptiSched: Uncertainty-Aware Enrollment Forecasting with CP-SAT Course Scheduling

Rooms sized to calibrated next-term demand bounds and scheduled conflict-free with constraint programming

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1. Problem and Approach

Rooms are assigned before registration opens. A room sized to last term's headcount or to a single forecast leaves students without seats when demand runs high.

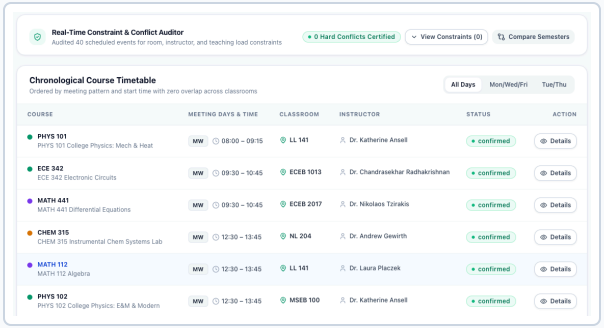
OptiSched forecasts next-term enrollment for each course and attaches calibrated 80% and 95% prediction intervals. Each room is sized to the 80% upper bound. A CP-SAT model then assigns rooms, time slots and instructors with zero hard conflicts.

Key contributions

- Next-term enrollment forecasts with split-conformal prediction intervals.
- Room sizes from the interval passed directly into a CP-SAT timetable model.
- Held-out evaluation on a real term: Spring 2026, 40 courses.
- What-If re-solves for enrollment surges, instructor release and room changes, with minimal schedule change.

2. System Pipeline

1. **Data.** Public UIUC registrar records (FOIA). 22,991 sections become 800 course-term enrollments (40 courses, 20 terms).
2. **Forecast.** Seasonal naive, damped trend and pooled ridge models are scored by a rolling-origin backtest; the lowest-MAE model is used, with split-conformal 80% and 95% intervals.
3. **Schedule.** CP-SAT assigns each course a room, a slot and an instructor. Hard constraints: room capacity, room and instructor overlap, teaching load. Soft constraints: faculty day, slot and room preferences (weights 40/40/20).
4. **What-If.** A dry-run re-solve under a scenario. It reports metric changes and every course that moved.



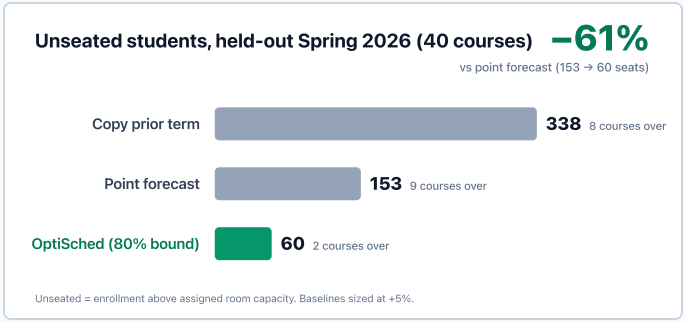
Master Schedule view. 40 sections; the auditor reports 0 hard conflicts.

Stack: FastAPI · OR-Tools CP-SAT · React + TypeScript

3. Data and Evaluation

- **Training data:** 19 terms, Fall 2016 to Fall 2025 (real). **Forecast target:** next term.
- **Test term:** Spring 2026, held out, 40 STEM courses (real enrollment).
- **Faculty preferences:** each instructor's most frequent slot over the prior 2 years (real, 25 of 40 courses).
- **Simulated:** room pool (16 rooms) and What-If scenarios.
- **Baselines:** copy prior term (Fall 2025) +5%, and point forecast +5%.
- **Unseated students:** enrollment above the assigned room capacity, summed over courses.

4. Results: Held-Out Spring 2026



39% **lower error than copy prior term**
MAE 17.24 vs 28.25; seasonal naive 18.03

85% **of courses inside the 80% interval**
34/40; 95% interval holds 39/40 (97.5%)

2 of 40 **courses overflowed their room**
vs 8 (prior term) and 9 (point forecast)

64.9% **seat utilization**
vs 64.5% in the actual UIUC schedule

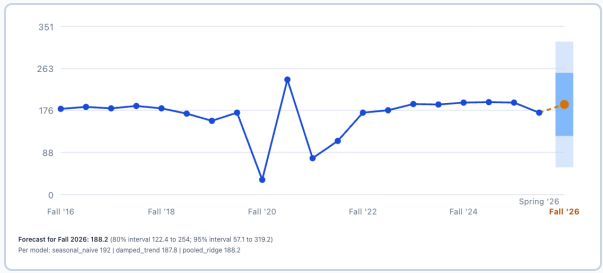
Schedule quality, Spring 2026	OptiSched	Actual UIUC
Empty-seat waste	35.1%	36.6%
Share of classes 10am to 2pm	22.5%	47.9%
Habit continuity (2-year)	84% (21/25)	n/a
Hard conflicts (OPTIMAL)	0	n/a

Who benefits. Students face fewer first-week seat shortages. Administrators make fewer room changes after registration, with no loss in seat utilization.

Code and Reproduction

github.com/JoeDoan/CS5551_Advanced-Software-Engineering-
Setup, demo and benchmark: see the README.

5. Forecast in the App

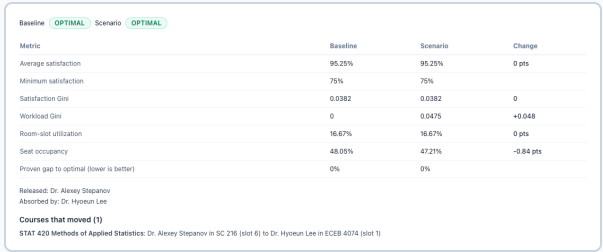


Demand Forecast view. STAT 400 history with the next-term forecast and 80%/95% intervals.

6. Working Demonstration: What-If

Each preset re-solves a simulated Spring 2027 term as a dry run. Placements the scenario does not force to change stay where they are.





What-If view, instructor-unavailable preset. Only STAT 420 moved.

Scenario and input	Output
Enrollment surge CS +20%, STAT +15%	2 courses move to larger rooms; both solves OPTIMAL
Instructor unavailable 1-course grant buyout	1 course moves to a qualified colleague; both solves OPTIMAL
Room reassigned All room capacities × 0.9	3 courses move to feasible rooms; both solves OPTIMAL

7. Extension: Faculty Commercialization Advisor

Drafts a UM-16C invention disclosure from an uploaded paper and retrieves clauses from 425 passages in 7 UM System and UMKC policy sources. Links to the instructor-release What-If to check teaching coverage.